



**Year 11 Mathematics Extension 1
Assessment Task 2
Term 3, Week 4, 2023**

Name: _____

Monday, 7th August 2023

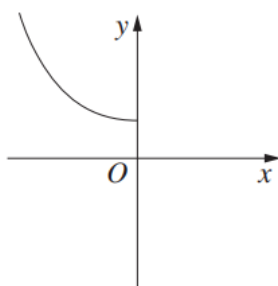
- Start each question on a new page.
- NESA approved calculators are permitted in this task.
- Marks may be deducted for insufficient or illegible work.
- Total possible mark: **32**
- Time allowed: **45 minutes + 2 minutes reading time**

Question 1 (7 marks)

Marks

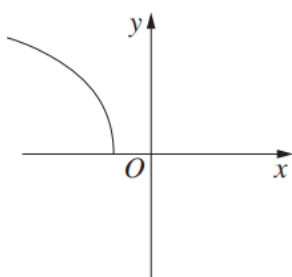
- (a) The diagram shows the graph $y = f(x)$.

1

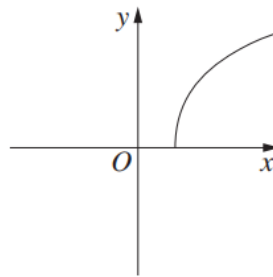


Which diagram shows the graph $y = f^{-1}(x)$?

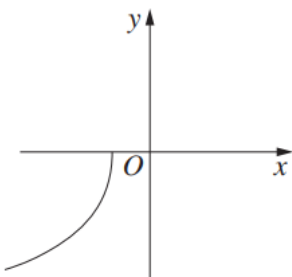
(A)



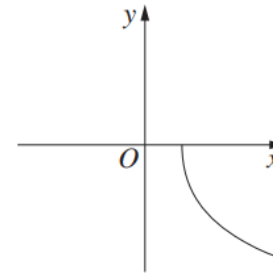
(B)



(C)



(D)



Question 1 continues on the next page

- (b) Show that $2 \cos\left(x + \frac{\pi}{6}\right) = \sqrt{3} \cos x - \sin x$. **1**
- (c) Solve $4^x - 2^{x+2} = 32$. **2**
- (d) Find the equation of the normal to the curve $y = xe^{2x}$ at the point where $x = \frac{\pi}{2}$. **3**

End of Question 1

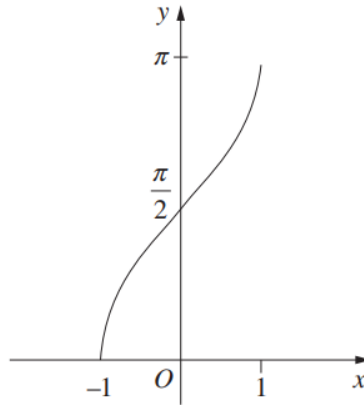
Question 2 starts on the next page

Question 2 (7 marks) *Start a new page*

Marks

- (a) The diagram shows the graph of a function.

1



Which function does the graph represent?

- (A) $y = \cos^{-1} x$
- (B) $y = \frac{\pi}{2} + \sin^{-1} x$
- (C) $y = -\cos^{-1} x$
- (D) $y = -\frac{\pi}{2} - \sin^{-1} x$
- (b) After time t years from the start of the year 2023 in Icebox, the number of people in the population is given by:

$$N = 70 + Ae^{0.1t}, \text{ where } A > 0.$$

- (i) Show that $\frac{dN}{dt} = 0.1(N - 70)$. **1**
- (ii) There were 100 people in the population at the start of the year 2023. Find the first year where all months will have a population size that exceeds 190. **2**

Question 2 continues on the next page

- (c) A piecewise function is defined by:

$$f(x) = \begin{cases} x^2 + 2x, & \text{for } x \leq 1 \\ -\frac{2}{2x-1} + 5, & \text{for } x > 1 \end{cases}$$

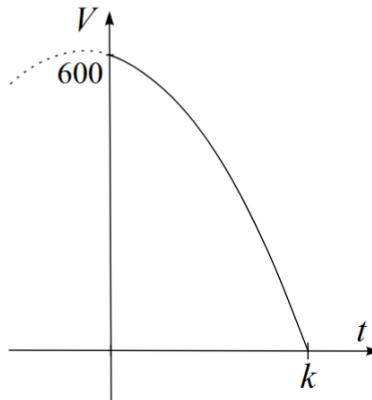
Providing justification for your answer:

- (i) Determine if $f(x)$ is continuous at $x = 1$. **1**
- (ii) Determine if $f(x)$ is differentiable at $x = 1$. **2**

End of Question 2

Question 3 starts on the next page

(a)



The diagram above represents the volume of water in a water tank as it is being drained. The volume of water in the tank is initially 600 Litres (L), and it take k minutes for the water to completely drain from the tank. It is known that the volume of water in litres after t minutes is given by $V = -30t^2 - 30t + 600$ for $0 \leq t \leq k$.

- (i) Find the value of k . **1**
- (ii) Find the average rate of change of the volume of water over the time that the tank is drained. **1**
- (iii) Find the time at which the instantaneous rate of change in volume, $\frac{dV}{dt}$, is equal to the average rate of change of the volume for the time it takes the tank to drain. **2**

- (b) Solve the pair of simultaneous equations: **2**

$$\log_2 x + \log_2 y = 3$$

$$\log_x y = 2$$

End of Question 3

Question 4 starts on the next page

Question 4 (6 marks) *Start a new page***Marks**

- (a) The length of a rectangle remains twice as its width at all times when the rectangle is expanding. The area of the rectangle is increasing at a rate of $16\text{ cm}^2/\text{sec}$. **3**

Find the rate at which the perimeter of the rectangle is increasing at the instant when its width is 200 cm.

- (b) Find the exact value of $\sin\left[\cos^{-1}\frac{3}{7} + \tan^{-1}\left(\frac{-3}{4}\right)\right]$. **3**

End of Question 4**Question 5 (6 marks)** *Start a new page***Marks**

- (a) Write the expression $\cos x + \cos 5x$ as the product of trigonometric ratios. **1**
- (b) (i) Show that $\tan(\alpha + \beta + \gamma) = \frac{\tan \alpha + \tan \beta + \tan \gamma - \tan \alpha \tan \beta \tan \gamma}{1 - \tan \alpha \tan \beta - \tan \alpha \tan \gamma - \tan \beta \tan \gamma}$. **3**
- (ii) If $\tan \alpha, \tan \beta, \tan \gamma$ are the roots of the equation $x^3 - (a+1)x^2 + (b-a)x - b = 0$, **2**
hence show that $\tan(\alpha + \beta + \gamma) = 1$.

End of Assessment



2023 Year 11 Mathematics Extension 1 Task 2 SOLUTIONS

Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
<u>QUESTION 1</u> a) Answer = 0. b) CHS: $2\cos(x + \pi/6)$ $2(\cos x \cos \frac{\pi}{6} - \sin x \sin \frac{\pi}{6})$ $2(\cos x \times (\frac{\sqrt{3}}{2}) - \sin x \times (\frac{1}{2}))$ $\sqrt{3}\cos x - \sin x$ c) $(2^2)^n - (2^2)(2^n) - 32 = 0$ let $u = 2^n$ $u^2 - 4u - 32 = 0$ $(u - 8)(u + 4) = 0$ $u = 8 \quad u = -4$ $2^n = 8 \quad 2^n = -4$ $\boxed{n = 3} \quad \text{no soln.}$ d) $y = u \cdot e^{2x}$ $u = u \mid v = e^{2x}$ $u' = 1 \mid v' = 2e^{2x}$ $y' = 2ue^{2x} + e^{2x}$ $y' = e^{2x}(2u + 1)$ at $x = \frac{\pi}{2}$ $y' = e^{\pi}(\pi + 1)$ $\therefore m_1 = e^{\pi}(\pi + 1)$ at $x = \frac{\pi}{2}$.	① correct expansion & sub of exact values ① correct sub & factorise ① soln ① for not stating no soln. ① correct dy/dx	normal: $m_1 \times m_2 = -1$ $m_2 = -\frac{1}{e^{\pi}(\pi + 1)}$ when $x = \frac{\pi}{2} \Rightarrow$ sub into y $y = \frac{\pi}{2}e^{\pi} \quad (\frac{\pi}{2}, \frac{\pi}{2}e^{\pi})$ eqn of normal $y - \frac{\pi}{2}e^{\pi} = -\frac{1}{e^{\pi}(\pi + 1)}(x - \frac{\pi}{2})$ <u>QUESTION 2</u> a) Answer: B. b) i) $N = 70 + Ae^{0.1t}$ $\frac{dN}{dt} = 0.1Ae^{0.1t}$ $= 0.1(N - 70)$ ii) when $t = 0, N = 600$ $600 = 70 + Ae^{0.1 \times 0}$ $\boxed{30 = A}$ let $N = 190$ $190 = 70 + 30e^{0.1t}$ $120 = 30e^{0.1t}$ $4 = e^{0.1t}$ $t = \frac{\ln 4}{0.1}$ $\approx 13.8629 \text{ years} \Rightarrow 14 \text{ years}$ $\therefore$ The year will be 2037.	① normal ① eqn of normal ① working & answer ① finding A ① t years $\rightarrow$ need to find year

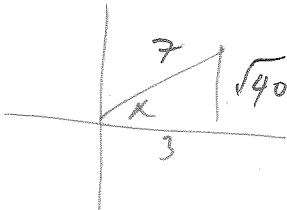
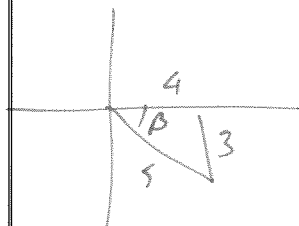


2023 Year 11 Mathematics Extension 1 Task 2 SOLUTIONS

Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
c) i) $f(1) = (1)^2 + 2 \times 1 = 3$. As $x \rightarrow 1^+$, $f(x) = \frac{-2}{2(1)-1} + 5 = 3$ $\therefore f(x)$ is continuous at $x=1$ $\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^+} f(x) = f(1)$ ii) $\frac{d}{dx}(x^2 + 2x) = 2x + 2$ When $x=1 \Rightarrow 2(1) + 2 = 4$ $\frac{d}{dx}\left(-\frac{2}{2x-1} + 5\right) = -2(2x-1)^{-1} + 5$ $= 2(2x-1)^{-2} \times 2$ $= \frac{4}{(2x-1)^2}$ When $x=1 : \frac{4}{(2(1)-1)^2} = 4$ As $x \rightarrow 1^-$, $f'(x) \rightarrow 4$ " $x \rightarrow 1^+$, $f'(x) \rightarrow 4$	① need to show appropriate value of state $\lim_{x \rightarrow 1^-} = \lim_{x \rightarrow 1^+}$ ① correct derivative of both ① show both limits are equal	<u>QUESTION 3</u> a) i) let $v=0$ $-30t^2 - 30t + 600 = 0$ $t^2 + t - 20 = 0$ $(t+5)(t-4) = 0$ $t = -5, t = 4$ but $t > 0$. $\therefore t = 4$ as $t > 0$. ii) Average rate of change $= \frac{-600}{4} \Rightarrow -150 \text{ L/min}$ iii) $\frac{dv}{dt} = -60t - 30$ $-150 \text{ L} = -60t - 30$ $-60t = -120$ $t = 2$. b) $\log_2 y = 2 \Rightarrow y = 2^2$ $\log_2 x + \log_2 y = 3$ ① $\log_2 xy = 3$ sub $y = 2^2$ $\log_2 x^3 = 3$ $x^3 = 2^3$ $x^3 = 8$ $x = 2$ / $y = 4$	① all correct ① answer ① correct $\frac{dv}{dt}$ ① answer ① use of any log laws correctly ① solve



2023 Year 11 Mathematics Extension 1 Task 2 SOLUTIONS

Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
<u>QUESTION 4</u> a) i)  $A = 2n^2 \Rightarrow \frac{dA}{dn} = 4n$ $\frac{dA}{dt} \Rightarrow 4n \times \frac{dn}{dt} = 16$ $n \frac{dn}{dt} = 4 \quad \Bigg \quad \frac{dn}{dt} = \frac{4}{n}$ When $n = 200$ cm $\frac{dn}{dt} = \frac{4}{200} \Rightarrow \frac{1}{50} \text{ cm/sec}$ $P = 6n$ $\frac{dP}{dt} = 6 \times 1 \times \frac{dn}{dt}$ $= 6 \times 1 \times \frac{1}{50}$ $= \frac{3}{25} \text{ cm/sec}$	① rate of change ① sub $n=200$ ① solving $\frac{dP}{dt}$	b) let $\alpha = \cos^{-1} \frac{3}{7}$ $\cos \alpha = \frac{3}{7} \cdot 0 \leq \alpha \leq \pi$  $\beta = \tan^{-1} \left(\frac{-3}{4} \right)$ $\tan \beta = \frac{-3}{4}, \quad -\frac{\pi}{2} < \beta < \frac{\pi}{2}$  $\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$ $= \frac{\sqrt{40}}{7} \times \frac{4}{5} + \frac{3}{7} \times \frac{-3}{5}$ ① answer $= \frac{4\sqrt{40}}{35} - \frac{9}{35}$ $= \frac{4\sqrt{40} - 9}{35}$	① finding $\cos \alpha$ ① finding $\tan \beta$



2023 Year 11 Mathematics Extension 1 Task 2 SOLUTIONS

Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
<u>Question 5</u> a) $\cos x + \cos 3x$ $= 2 \cos 2x \cos x$ when $A+B = 5$ $A-B = 1$ b) i) $\tan(x+B+\theta) = \frac{\tan x + \tan(B+\theta)}{1 - \tan x \tan(B+\theta)}$ $\tan a + \left(\frac{\tan B + \tan \theta}{1 - \tan B \tan \theta} \right) \times (1 - \tan B \tan \theta)$ $\frac{1 - \tan a \left(\frac{\tan B + \tan \theta}{1 - \tan B \tan \theta} \right) \times (1 - \tan B \tan \theta)}{\tan a (1 - \tan B \tan \theta) + \tan B + \tan \theta}$ $\frac{1(1 - \tan B \tan \theta) - \tan a (\tan B + \tan \theta)}{\tan a - \tan a \tan B \tan \theta + \tan B + \tan \theta}$ $\frac{1 - \tan B \tan \theta - \tan a \tan B - \tan a \tan \theta}{\tan a + \tan B + \tan \theta - \tan a \tan B \tan \theta}$ $\frac{1 - \tan a \tan B - \tan B \tan \theta - \tan a \tan \theta}{\tan a + \tan B + \tan \theta}$ ① working ① answer	① answer	ii) <u>sum of roots</u> : $\tan x + \tan B + \tan \theta = \frac{-b}{a}$ $= a - 1$ <u>product of pair sum</u> $\tan x \tan B + \tan B \tan \theta + \tan x \tan \theta = \frac{c}{a}$ $= b - a$ <u>product of roots</u> $\tan x \tan B \tan \theta = \frac{-d}{a}$ $= b$ from part (i) $\tan(x+B+\theta) = \frac{a+1-b}{1-(b-a)}$ $= \frac{a+1-b}{1+a-b}$ $= 1$	① for all 3. ① correct sub